

Mohammad Elayan, MSc

Graduate Research & Teaching Assistant

🏢 Dept. of Civil & Environmental Engineering

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Profile

PhD student of the Civil & Environmental Engineering at the University of Nebraska–Lincoln. Research focuses on intelligent, human-aware autonomous driving in mixed traffic, with an emphasis on balancing competing driving objectives and modeling driver heterogeneity. Nearly a decade of transportation engineering experience in Dubai, leading numerous projects across the MENA region.

Education

#	Degree — Program & Institution	Grade	Year
1	PhD — Civil Engineering (Transportation Engineering) <i>University of Nebraska-Lincoln, Lincoln, NE, USA</i>	Ongoing	2023–Present
2	MSc — Civil Engineering (Transportation Engineering) <i>Jordan University of Science & Technology, Irbid, Jordan</i>	91.6% · Rank 1	2011–2013
3	BSc — Civil Engineering (Transportation Engineering) <i>Jordan University of Science & Technology, Irbid, Jordan</i>	82.3% · Rank 16	2006–2011

Research

Peer-Reviewed Publications

(7)

1. **Elayan, M.**, & Kontar, W. (2026). "Can Automated Vehicles Have it All? A Consensus Framework for Diagnosing Behavioral Trade-Offs in Mixed Traffic." *Transportation Research Part C: Emerging Technologies*. [Link]
2. **Elayan M.**, & Kontar, W. (2026). "Consensus-Aware AV Behavior: Trade-Offs Between Safety, Interaction, and Performance in Mixed Urban Traffic." *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, Gold Coast, Australia. [Link]
3. **Elayan, M.**, Karki, S., & Hawkins, J. (2025). "Better Safety Analyses through Smarter Data: Adding Open-Street-View and Traffic Calibrated-LBS Data to Pedestrian Crash Analysis in Lincoln, NE." *Transportation Research Record*. [Link]
4. **Elayan, M.**, Aldridge, N., Hawkins, J., & Nam, Y. (2025). "Integrating StreetLight, EPS Smart Location Data and Road Attributes: A Random Forest Approach to Multi-Modal Traffic Calibration in Lincoln, Nebraska." *Journal of Transportation Engineering, Part A: Systems*, Vol. 151(8). [Link]
5. Al-Khateeb, G., Al-Suleiman, T., Khedaywi, T., & **Elayan, M.** (2016). "Studying Rutting Performance of Superpave Mixtures." *Road Materials and Pavement Design*, Vol. 19(2). [Link]
6. Alsheyab, M., Khedaywi, T., & **Elayan, M.** (2013). "Laboratory Study on Solidification/Stabilization of Unwanted Medications Using Asphalt as a Binder." *Journal of Material Cycles and Waste Management*, Vol. 15(2). [Link]
7. Al-Suleiman, T., & **Elayan, M.** (2013). "Gap Acceptance Behavior at U-turn Median Openings – Case Study in Jordan." *Jordan Journal of Civil Engineering*, Vol. 7(3). [Link]

I Other Publications

(5)

1. Armantalab, O., **Elayan, M.**, Zhao, L., Huynh, N., & Kontar, W. (2026). 'A Quantum-Inspired Representation of Dynamic Travel Behavior.' [Link]
2. **Elayan, M.**, & Kontar, W. (2026). "Behavioral Heterogeneity as Quantum-Inspired Representation." *arXiv preprint arXiv:2603.22729*. [Link]
3. **Elayan, M.**, & Kontar, W. (2026). "Learning the Pareto Space of Multi-Objective Autonomous Driving: A Modular, Data-Driven Approach." *arXiv preprint arXiv:2601.18913*. [Link]
4. **Elayan, M.**, & Kontar, W. (2025). "Consensus-Aware AV Behavior: Trade-offs Between Safety, Interaction, and Performance in Mixed Urban Traffic." *arXiv preprint arXiv:2505.04379*. [Link]
5. Nam, Y., Hawkins, J., Butler, D., Aldridge, N., **Elayan, M.**, & Yoo, J. (2024). "Modeling Pedestrian and Bicyclist Crash Exposure with Location-Based Service Data." *Nebraska Dept. of Transportation, Technical Report No. SPR-FY23(025)*. [Link]

I Conference Papers & Presentations

(11)

1. Gabrys, G., **Elayan, M.**, Armantalab, O., & Kontar, W. (2026). "Where Pharmaceutical Deserts Are and How to Serve Them: A Geospatial Index and Autonomous Routing Study in Rural Nebraska." *TRB 106th Annual Meeting*, Washington, DC. [Accepted]
2. **Elayan, M.**, & Kontar, W. (2026). "Learning Behavioral Consensus for Autonomous Driving in Mixed Traffic." *TRB 106th Annual Meeting*, Washington, DC. [Accepted]
3. **Elayan, M.**, Armantalab, O., & Kontar, W. (2026). "Quantum-Inspired Modeling of Driving Behavior." *TRB 106th Annual Meeting*, Washington, DC. [Accepted]
4. **Elayan, M.**, & Kontar, W. (2026). "Behavioral Heterogeneity as Quantum-Inspired Representation." *IEEE ITSC*, Naples, Italy. [Presented]
5. **Elayan, M.**, & Kontar, W. (2026). "Learning the Pareto Space of Multi-Objective Autonomous Driving: A Modular, Data-Driven Approach." *IEEE Intelligent Vehicles Symposium (IV)*, Detroit, MI. [Presented]
6. **Elayan, M.**, & Kontar, W. (2026). "Consensus-Aware AV Behavior: Empirical Pareto-Guided Reinforcement Learning for Mixed Traffic." *NSF NAIRR Annual Meeting*, Arlington, VA. [Presented]
7. **Elayan, M.**, & Kontar, W. (2026). "The Empirical Pareto Frontier of Automated Driving." *College of Engineering 7th Annual Graduate Student Symposium*, UNL. [Presented]
8. **Elayan, M.**, & Kontar, W. (2026). "A Unified Framework for Consensus-Aware AV Behavior." *TRB 105th Annual Meeting*, Washington, DC. [Presented]
9. Armantalab, O., **Elayan, M.**, & Hawkins, J. (2025). "Multi-Stage Clustering of Daily Activity Time Series." *TRB 105th Annual Meeting*, Washington, DC. [Presented]
10. **Elayan, M.**, & Kontar, W. (2025). "Consensus-Aware AV Behavior: Trade-offs Between Safety, Interaction, and Performance in Mixed Urban Traffic." *IEEE ITSC*, Gold Coast, Australia. [Presented]
11. **Elayan, M.**, Karki, S., & Hawkins, J. (2025). "Better Safety Analyses through Smarter Data." *TRB 104th Annual Meeting*, Washington, DC. [Presented]

I Under Review

(3)

1. **Elayan, M.**, Armantalab, O., & Kontar, W. (2026). "Quantum-Inspired Modeling of Driving Behavior." *Transportation Research Part B: Methodological (Special Issue: Mathematical Foundations for Trustworthy AI Applications in Transportation Systems)*.
2. Armantalab, O., **Elayan, M.**, Zhao, L., Huynh, N., & Kontar, W. (2026). 'A Quantum-Inspired Representation of Dynamic Travel Behavior.' *Transportation Research Part C: Emerging Technologies*.
3. Armantalab, O., **Elayan, M.**, & Hawkins, J. (2026). "Multi-Stage Clustering of Daily Activity Time Series: Patterns and Socio-Demographic Insights." *Transportmetrica A: Transport Science*.

I Research Projects

-  Modeling Pedestrian and Bicyclist Crash Exposure with Location-Based Service Data.

Nebraska Department of Transportation, USA.

2022–2024

 *Introducing New Technologies and Advanced Methodologies for Asphalt Mixtures of Highway Pavements in Jordan.*

Ministry of Higher Education and Scientific Research, Amman, Jordan.

2008–2012

Grants & Awards

- ★ NSF National AI Research Resource (NAIRR) Pilot Grant 2025
“Consensus-Seeking Autonomous Agents in Complex Cyber-Physical Transportation Systems” (Award No. NAIRR250132).
National Science Foundation, USA. Role: Researcher.

Fellowships & Scholarships

- 🏆 University of Nebraska-Lincoln College of Engineering Professional Development Fellowship 2025, 2026
🏆 Patrick T. McCoy Fund for Engineering Excellence 2025, 2026
🏆 MATC & NTC Travel Fund Scholarship 2024, 2025, 2026

Experience

Graduate Research & Teaching Assistant

Jan 2023 – Present

University of Nebraska-Lincoln, Lincoln, NE, USA

Member of Dr. Wissam Kontar’s research team at the Civil & Environmental Engineering Department and Nebraska Transportation Center. Research focuses on traffic agent heterogeneity and automated vehicle behavior control.

Teaching Assistantships:

- ▶ CIVE 202 – Civil Engineering Analysis II (programming in civil engineering)
- ▶ CIVE 361 – Principles of Transportation Engineering

Senior Transportation Engineer

Jun 2013 – Jan 2023

TrafQuest, Dubai, United Arab Emirates

Roles encompassed travel demand forecasting, traffic operations, multi-modal transport evaluation, microsimulation, safety analysis, geometric design, and technical reporting. Contributed to **100+ projects** across UAE and MENA, serving as Project Manager for the majority.

Project Category	Total	Engineer	Manager
Transportation Master Plan	20	12	8
Traffic Impact Study	64	24	40
Transportation Integration Study	11	1	10
Strategic Freight Transport Study	7	7	0
Parking Study	3	1	2
Traffic Management Plan	3	1	2
Evacuation Plan	1	0	1
Total	109	46	63

Graduate Research & Teaching Assistant

2011 – 2013

Jordan University of Science and Technology, Irbid, Jordan

Research in traffic operations, highway materials, and waste management: data collection, laboratory experiments, statistical analysis, and technical writing. Teaching: Surveying Lab, Highway Materials Lab, and Engineering Statics.

Teaching

CIVE 202 – Civil Engineering Analysis II

2024 – Present

University of Nebraska-Lincoln, Lincoln, NE, USA

A project-based course teaching civil engineers to program in Python, enabling them to collect, organize, analyze, and visualize data, and automate engineering calculations. Pairs core concepts from mechanics, dynamics, and sustainable design with professional programming skills, so students learn to translate a client's problem into a working solution.

CIVE 361 – Principles of Transportation Engineering

2024 – Present

University of Nebraska-Lincoln, Lincoln, NE, USA

Introduces transportation engineering through operator, user, and vehicle characteristics, transportation facility design, traffic operations and control, and transportation planning. Draws on Mannering's *Principles of Highway Engineering and Traffic Analysis* alongside real trajectory and traffic datasets from ongoing research.

CE 343 – Surveying Laboratory

2012 – 2013

Jordan University of Science and Technology, Irbid, Jordan

A hands-on lab progressing from taping and chain surveying through levels, theodolites, tacheometers, EDM, and the total station, closing with field projects in traversing, contouring, and area measurement by planimeter.

CE 201 – Engineering Statics

2012 – 2013

Jordan University of Science and Technology, Irbid, Jordan

A foundational mechanics course spanning vector analysis, statics of particles, equivalent force systems on rigid bodies, equilibrium, supports and reactions, truss analysis by joints and sections, frames, shear and bending moment diagrams, and centroids and moments of inertia of composite areas.

Mentorship

Undergraduate Research Mentorship

Grace Gabrys

Jun – Aug 2026

NSF Research Experiences for Undergraduates (REU) Program

University of Nebraska-Lincoln, Lincoln, NE, USA.

Supervised research focusing on the geospatial indexing of pharmaceutical deserts in Nebraska and optimization-based analysis of conventional and autonomous delivery options.

Mawunyo Tchamdja

Jun – Aug 2025

NSF Research Experiences for Undergraduates (REU) Program

University of Nebraska-Lincoln, Lincoln, NE, USA.

Supervised research focusing on autonomous electric mobility systems in Rural Nebraska as part of the program.

Service

Peer Review

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|  <i>Transportation Research Part A: Policy and Practice</i> | ×2 |
|  <i>KSCE Journal of Civil Engineering</i> | ×1 |
|  <i>Transportation Research Board (TRB) Annual Meeting (2027)</i> | ×4 |
|  <i>Discover Civil Engineering (Springer Nature)</i> | ×1 |

Skills

Areas of Expertise

Automated Driving Traffic Microsimulation
 Reinforcement Learning Machine Learning
 Statistical Analysis Optimization
 Traffic Safety Analysis Crash Analysis
 Behavior Heterogeneity Transport Modeling
 Demand Forecast Traffic Analysis
 Systems Accessibility Soft Mobility
 Transport Integration Transport Planning
 Geometric Design

Software & Programming

Languages

Python Julia R SPSS SAS

Simulation & Planning

SUMO VISSIM VISUM Synchro

SIDRA HCS

GIS

ArcGIS QGIS GeoDa

Other

MS Office $\LaTeX$ HTML Photoshop Lightroom

References

Name	Role	Organisation	E-mail
Wissam Kontar	Assistant Professor	University of Nebraska-Lincoln	wkontar2@nebraska.edu
Jason Hawkins	Assistant Professor	University of Calgary	jfhawkin@ucalgary.ca
Ghazi Al-Khateeb	Professor / Chairman	University of Sharjah	galkhateeb@sharjah.ac.ae
Mohammad Braiwish	Managing Director	TrafQuest	m.younis@trafquest.com